

TIØ4145 Corporate Finance

Exercise 6: Capital structure – Part 2

Deadline: Monday October 13th, 2025, 23:59 – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday October 9th, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

Relevant Curriculum

- Lecture notes from lecture given on October 6th.
- Berk and DeMarzo, Corporate Finance 5th edition: Chapters 16 and 17.

Tasks

Chapter 16 – Financial Distress, Managerial Incentives, and Information

1. Kohwe Corporation plans to issue equity to raise \$40 million to finance a new investment. After making the investment, Kohwe expects to earn free cash flows of \$10 million each year. Kohwe currently has 5 million shares outstanding, and it has no other assets or opportunities. Suppose the appropriate discount rate for Kohwe's future free cash flows is 8%, and the only capital market imperfections are corporate taxes and financial distress costs.
 - a. What is the NPV of Kohwe's investment?
 - b. Given these plans, what is Kohwe's value per share today?

Suppose Kohwe borrows the \$40 million instead. The firm will pay interest only on this loan each year, and it will maintain an outstanding balance of \$40 million on the loan. Suppose that Kohwe's corporate tax rate is 40%, and expected free cash flows are still \$10 million each year.

- c. What is Kohwe's share price today if the investment is financed with debt?

Now suppose that with leverage, Kohwe's expected free cash flows will decline to \$9 million per year due to reduced sales and other financial distress costs. Assume that the appropriate discount rate for Kohwe's future free cash flows is still 8%.

- d. What is Kohwe's share price today given the financial distress costs of leverage?

2. Baruk Industries has no cash and a debt obligation of \$36 million that is now due. The market value of Baruk's assets is \$81 million, and the firm has no other liabilities. Assume perfect capital markets.

- a. Suppose Baruk has 10 million shares outstanding. What is Baruk's current share price?
- b. How many new shares must Baruk issue to raise the capital needed to pay its debt obligation?
- c. After repaying the debt, what will Baruk's share price be?

3. Info Systems Technology (IST) manufactures microprocessor chips for use in appliances and other applications. IST has no debt and 200 million shares outstanding. The correct price for these shares is either \$15.50 or \$13.50 per share. Investors view both possibilities as equally likely, so the shares currently trade for \$14.50.

IST must raise \$550 million to build a new production facility. Because the firm would suffer a large loss of both customers and engineering talent in the event of financial distress, managers believe that if IST borrows the \$550 million, the present value of financial distress costs will exceed any tax benefits by \$15 million. At the same time, because investors believe that managers know the correct share price, IST faces a lemons problem if it attempts to raise the \$550 million by issuing equity.

- a. Suppose that if IST issues equity, the share price will remain at \$14.50. To maximize the long-term share price of the firm once its true value is known, would managers choose to issue equity or borrow the \$550 million if:
- They know the correct value of the shares is \$13.50?
 - They know the correct value of the shares is \$15.50?
- b. Given your answer to part (a), what should investors conclude if IST issues equity? What will happen to the share price?

- c. Given your answer to part (a), what should investors conclude if IST issues debt? What will happen to the share price in that case?
 - d. How would your answers change if there were no distress costs, but only tax benefits of leverage?
4. Ralston Enterprises has assets that will have a market value in one year as follows:

Probability	1%	6%	24%	38%	24%	6%	1%
Value (\$ million)	70	80	90	100	110	120	130

That is, there is a 1% chance the assets will be worth \$70 million, a 6% chance the assets will be worth \$80 million, and so on. Suppose the CEO is contemplating a decision that will benefit her personally but will reduce the value of the firm's assets by \$10 million. The CEO is likely to proceed with this decision unless it substantially increases the firm's risk of bankruptcy.

- a. If Ralston has debt due of \$75 million in one year, the CEO's decision will increase the probability of bankruptcy by what percentage?
- b. What level of debt provides the CEO with the biggest incentive not to proceed with the decision?

Chapter 17 – Payout Policy

1. AMC Corporation currently has an enterprise value of \$400 million and \$100 million in excess cash. The firm has 10 million shares outstanding and no debt. Suppose AMC uses its excess cash to repurchase shares. After the share repurchase, news will come out that will change AMC's enterprise value to either \$600 million or \$200 million.
 - a. What is AMC's share price prior to the share repurchase?
 - b. What is AMC's share price after the repurchase if its enterprise value goes up? What is AMC's share price after the repurchase if its enterprise value declines?
 - c. Suppose AMC waits until after the news comes out to do the share repurchase. What is AMC's share price after the repurchase if its enterprise value goes up? What is AMC's share price after the repurchase if its enterprise value declines?

- d. Suppose AMC management expects good news to come out. Based on your answers to parts b and c, if management desires to maximize AMC's ultimate share price, will they undertake the repurchase before or after the news comes out? When would management undertake the repurchase if they expect bad news to come out?
 - e. Given your answer to part d, what effect would you expect an announcement of a share repurchase to have on the stock price? Why?
2. Suppose the stock of Host Hotels & Resorts is currently trading for \$20 per share.
 - a. If Host issued a 20% stock dividend, what will its new share price be?
 - b. If Host does a 3:2 stock split, what will its new share price be?
 - c. If Host does a 1:3 reverse split, what will its new share price be?
3. [Voluntary]
 EJH Company has a market capitalization of \$1 billion and 20 million shares outstanding. It plans to distribute \$100 million through an open market repurchase. Assuming perfect capital markets:
 - a. What will the price per share of EJH be right before the repurchase?
 - b. How many shares will be repurchased?
 - c. What will the price per share of EJH be right after the repurchase?

Exam Question – December 2015

Firm XYZ has assets with a market value of \$1 billion, \$100 million of which are cash. The firm has 10 million shares outstanding and debt of \$300 million. Assume perfect capital markets.

- a. What is the firm's current stock price?

The firm plans to distribute \$50 million as dividend.

- b. What would its share price be after the dividend is paid?
- c. What would its new market debt-equity ratio be after this transaction?

If the firm would distribute \$50 million through an open market repurchase instead.

- d. How many shares would it repurchase?
- e. What would the price per share of firm XYZ be right after the repurchase?

Guest lecture questions

- a. What challenges arise when a firm is in financial distress and heading towards bankruptcy?
- b. What are the different challenges equity holders vs. debt holders face in a bankruptcy?
- c. If a firm enters bankruptcy and the debt holders take over the company what is the most likely outcome? Why?
- d. What are some alternatives to bankruptcy and why are they sometimes more attractive?

References

Berk, J. and DeMarzo, P. (2017). Corporate Finance, Fifth Edition, Pearson Education, Essex, England.